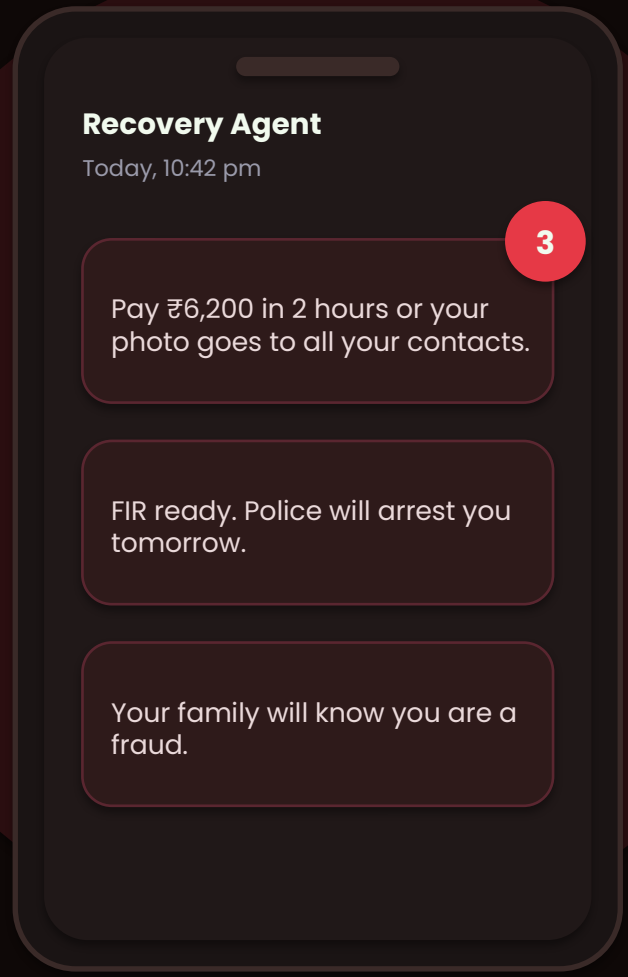
SOCIAL & SUSTAINABILITY PROJECT

Digital Loan App Harassment in India

Framing the crisis through the UN SDGs, and our solution: LoanShield.



1 8 10 16 Four SDGs, one problem

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THE CRISIS

Illegal Digital Lending Traps Borrowers

Quick Loans Attract the Vulnerable → **Hidden Fees Deepen Debt** → **Aggressive Recovery Tactics**

The harm spreads beyond the borrower — affecting families, workplaces, and entire community trust in digital finance.

WHY IT MATTERS

Harassment Tactics & Real-World Harm

Threatening Calls & Messages

Abusive language and fake legal notices create constant fear.

Contact-List Blackmail

Personal data is weaponized to shame and pressure borrowers.

Fear, Stress & Family Conflict

Repeated harassment causes psychological harm and tears families apart.

Privacy Violations

Data misuse exposes victims and destroys trust in digital systems.

Access Without Protection

No Institutional Accountability

Unsustainable Digital Finance

1

No Poverty

Predatory Lending Creates a Poverty Trap

High Charges

Push vulnerable households deeper into debt

Debt Cycles

Forced to re-borrow just to survive

Essential Cuts

Less spending on food, health, education

Financial Ruin

Poverty becomes impossible to escape

Predatory lending doesn't just cause debt — it makes poverty structurally harder to escape, directly undermining SDG 1.

8

Decent Work & Economic Growth

10

Reduced Inequalities

Unsafe Lending Undermines Fair Economic Participation

SDG 8 — Livelihoods at Risk

Harassment calls at work disrupt employment. Constant stress reduces productivity. Predatory debt makes decent work meaningless.

SDG 10 — Inequality Widens

Vulnerable and marginalized groups face the highest risk of exploitation. Without safe credit, existing inequalities only grow.

16

Peace, Justice & Strong Institutions

Threats, Data Abuse & Eroding Trust

Threats & Fake Legal Notices

Create constant fear and insecurity, undermining personal safety and dignity.

Privacy Violations

Misuse of personal data weakens trust in digital institutions and financial systems.

No Accessible Justice

Victims lack clear pathways for accountability, leaving exploiters unchallenged.

Without strong institutions to regulate lending and protect data, digital finance becomes a tool of coercion.

ONE PROBLEM – FOUR SDGs

Digital finance must empower, not exploit.

- SDG 1 — Deepens poverty & debt traps
- SDG 8 — Disrupts livelihoods & work
- SDG 10 — Widens existing inequalities
- SDG 16 — Erodes trust & justice

One problem. Four SDGs. Loan-app harassment is not just a fintech issue — it is a sustainable development crisis. Safe, fair, and accountable digital finance is essential for inclusive growth in India.

PROBLEM RECAP

Where the problem leaves victims

Easy loan, hidden cost

₹5,000 promised, ₹3,500 paid, full repayment in 7 days.

→

Threats & blackmail

Contact lists, morphed photos and fake police notices.

→

Silence & rebrand

The victim pays and stays quiet. The app returns under a new name.

What is missing: a way for victims to fight back, together, with proof.

1 8 10 16

One problem, four SDGs. Our solution answers each stage of it.

OBJECTIVES

What our solution sets out to do

1 Calm the victim

Answer every threat with the facts and the law it breaks.

2 Build evidence automatically

Seal each message as tamper-proof proof.

3 Protect the contacts

Warn family and friends before the scammers reach them.

4 Unmask the operator

Link rebranded apps by UPI ID, phone number and wording.

5 Offer a fair alternative

Small, fair loans through SHGs and microfinance groups.

6 Reach everyone

Tamil, Hindi and English, WhatsApp for basic phones.

OUR SOLUTION

LoanShield

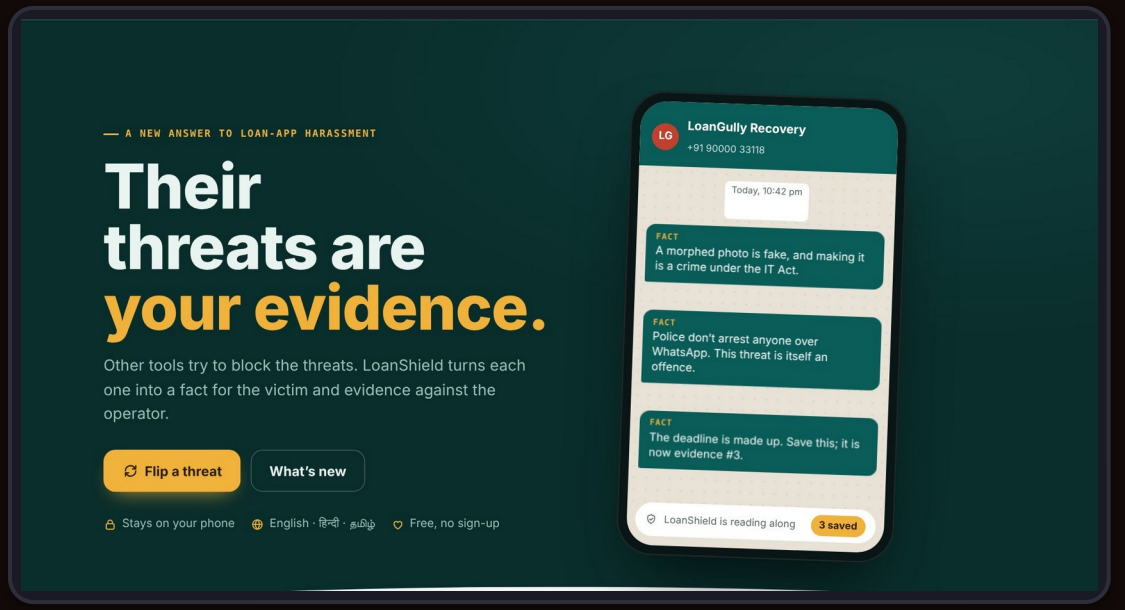
Their threats become evidence.

A community system with a website at its centre. Other tools try to block the threats. LoanShield turns each one into a fact for the victim and evidence against the operator.

Flip Every threat is answered with what is actually true.

Seal Every message becomes tamper-proof evidence.

Link Every UPI ID leads back to one operator.



Live: https://karthik705.github.io/loanshield/

HOW IT WORKS

Five stages, one loop

1 Prevent

COMMUNITY

- Real-cost camps at SHGs and colleges
- Fair small loans via SHGs and microfinance

→

2 Detect

WEBSITE

- Link reports by UPI ID, number, wording
- Public only after 3 people report

→

3 Protect

WEBSITE + NGO

- Flip each threat into fact and law
- Warn contacts in 3 languages

→

4 Respond

COMMUNITY

- Legal-aid clinics help file
- Leaked-photo takedowns by NGO counsellors

→

5 Reform

COMMUNITY

- Public dashboard of operators
- Reports to app stores, payment apps

↻ **Loop:** hotspots found in Reform decide where the next Prevent camps run.

■ Carried by the website ■ Carried by people and partners

THE WEBSITE: ONE PART OF THE SYSTEM

Paste a threat. Watch it flip.

Speak, paste or screenshot

Say what happened in Tamil, Hindi or English; the key details are pulled out.

Fear → Fact

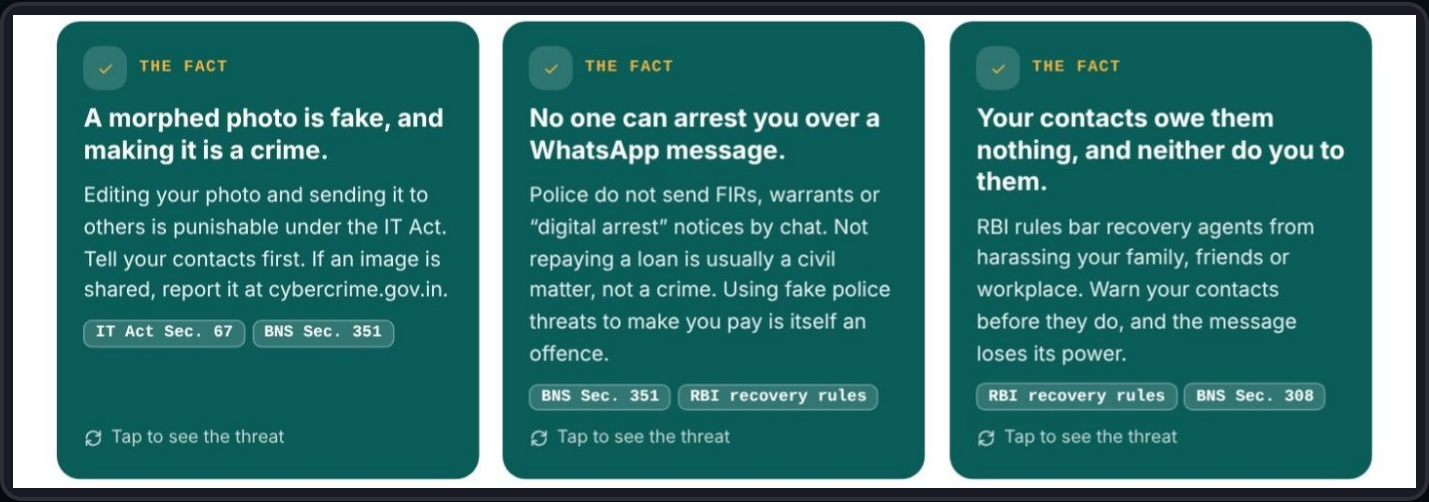
Each threat card turns over to show the truth and the law it breaks.

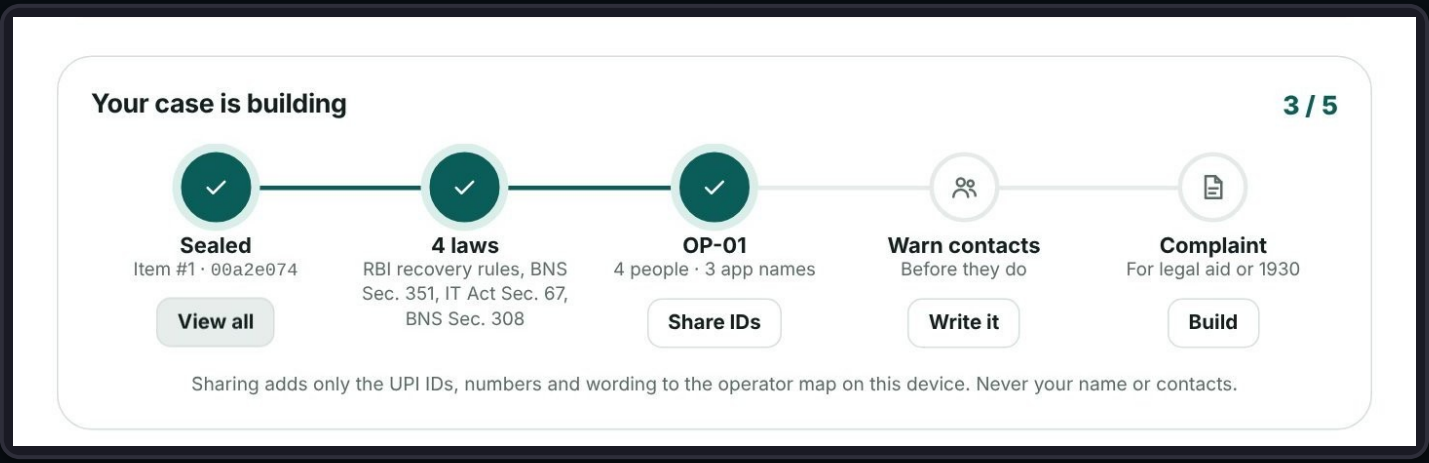
Protect me in one tap

Seals the message, links the operator, writes the warning and complaint.

Case tracker

Sealed evidence, laws, operator, contact warning, complaint.





TECH STACK

Simple tools, chosen to reach everyone

Website

Opens in any phone browser. No app to install.

HTML · CSS · JavaScript

Reading threats

Spots threat words in English, Hindi and Tamil.

Keyword detection + AI for screenshots

Voice

Speak instead of typing; hear the facts read out.

Phone’s built-in speech tools

Evidence seal

A digital fingerprint shows if anything was changed.

SHA-256 hashing

Linking operators

Matches UPI IDs and numbers across reports.

Pattern matching

Privacy & hosting

Checked on the phone; shared only by choice. Free to run.

GitHub Pages

FOR PEOPLE WHO STRUGGLE WITH TECH

Two ways in: do it yourself, or get assistance

Do it yourself

Paste, speak or upload a screenshot of the threat. About a minute.

Someone is with me

Helper mode for family or volunteers: big text, 4 simple steps, read aloud.

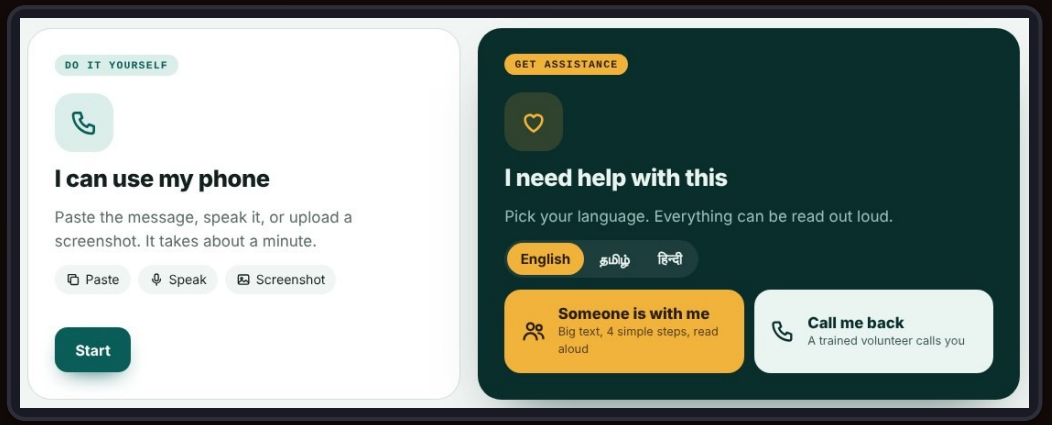
Call me back

Leave a number; a trained NGO volunteer calls in their language.

Their language, out loud

Tamil, Hindi or English with one tap; facts are spoken.

No smartphone at all? Student volunteers run weekly help desks at SHG meetings and do it for them.



The first choice on the website.

WHEN A MORPHED PHOTO LEAKS

Evidence is not enough. Dignity comes first.

Before it spreads

- Contact Shield warns family first: “any photo of me is fake”
- The photo arrives already discredited

Within hours

- Ready takedown request for each platform
- Platforms must act within 2 hours of a complaint (IT Rules, 2026)
- StopNCII fingerprint blocks re-uploads; the image never leaves the phone

For the person

- An NGO counsellor does the takedowns for them
- A woman volunteer on request
- They never have to see it again

On the website: “This photo is fake. Making it is the crime, not being in it.” One button opens the whole checklist.

NOVELTY

What is new in our approach

USUAL APPROACH	LOANSHIELD
Spam blockers hide the message. The victim is still scared.	**Flips fear into facts** — Each threat is answered with the truth and the law it breaks.
More threats mean more fear. Each victim complains alone.	**Every threat builds the case** — Messages are sealed as evidence; contacts are warned first.
One app is reported. It returns next week with a new name.	**Follows the money** — Reports sharing a UPI ID or number join into one operator.

Other tools defend one phone. LoanShield turns every victim’s report into one case against the operator.

SDG IMPACT

How the solution serves the SDGs

16

MAJOR SDG

Peace, Justice & Strong Institutions

- Access to justice: sealed evidence and legal-aid help to file
- Less violence and fear: threats answered with facts
- Accountability: operators named in public, not just apps
- Privacy protected: checked on the phone, shared only by consent

1

SDG 1 · No Poverty

Fair SHG loans break the debt trap; real-cost checks before borrowing.

8

SDG 8 · Decent Work

Contact Shield keeps threats from reaching employers.

10

SDG 10 · Reduced Inequalities

Local languages and WhatsApp reach low-literacy users.

FUNDING & PARTNERS

Run with cyber-safety NGOs

CyberPeace Foundation

Cyber-safety NGO. Pilot funding, counsellor network and victim helpdesk support.

Responsible Netism

Online-safety NGO. Awareness workshops in colleges and communities.

Cyber Society of India

Chennai-based cyber awareness body. Volunteer training for our Tamil Nadu pilot.

CSR grants: from responsible fintech companies and banks

Law-college clinics: pro-bono legal help for filing

SHGs & microfinance groups: the fair-loan alternative

Student volunteers: NSS and social clubs run camps

Partners we plan to approach for the pilot. Running cost is low: the website is free to host.

CHALLENGES & LIMITATIONS

What makes it hard, and what it can’t do yet

CHALLENGES → HOW WE HANDLE THEM

Apps rebrand in weeks
→ link by UPI ID and number, not app name

Shame stops reporting
→ private by default, NGO counsellors

False or spiteful reports
→ public only after 3 different people

Low digital literacy
→ WhatsApp, voice and local languages

LIMITATIONS

Keyword scanner can miss new wording and other languages

Tested on 50 messages written by our team, not field data

Evidence is stored only on the victim’s own device

Legal mapping is a guide, not legal advice

People-side stages depend on NGO partners

FUTURE SCOPE

Where LoanShield goes next

Android app

Share any message straight into LoanShield; log late-night calls automatically.

Smarter reading

A trained model for Tamil, Telugu and Bengali threats, and voice messages.

WhatsApp bot

Forward a threat to one number and get the facts back as a voice note.

Voice notes

Turn threatening voice messages into text evidence.

90-day pilot

One SHG federation and one cyber-safety NGO in Chennai.

Open data for NGOs

Share operator links with partner NGOs to speed up takedowns.

THANK YOU

Safer loans.

Stronger people.

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PASUNURU SAI VIVEK - 24BCE1239

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Try LoanShield

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